

Self-contained proof of Theorem 3.5

Notation and conventions as per the Notation document.

Structure

The proof of Theorem 3.5 in [1] rests on four lemmas. In this document I state them in the relevant form for the proof of Theorem 3.5 in [1]. I also give the four lemmas simple, meaningful names for ease of reference.

1. *reformulation (Lemma 2)*: rewrites $\{x_k\}$ and $\{y_k\}$ from NAG with a third sequence $\{z_k\}$ in a purely algebraic step that does not change the values of the original sequences.
2. *potential (Lemma 4)*: shows that a particular ‘potential function’ that involves the sub-optimality gap $f - f_\star$ and an ‘amortised cost’ related to $\|z_k - x_\star\|$ is monotonic and converges to a finite limit *when* $\{t_k\}$ satisfies the Nesterov quadratic relation (4).
3. *boundedness (Lemma 5)*: shows that $\{x_k\}$ is bounded.
4. *convergence (Lemma 7)*: shows that a sequence where a particular amplified version of the increments converges will converge.

In the next sections I construct a complete proof in a unified notation for the above lemmas, followed by the proof of Theorem 3.5 itself. Within reason, I have also attempted to include the statement and proof of any other results used in relevant sections. Many of these are standard tools, such as the Stolz-Cesàro Theorem, but were not known to me beforehand.

Introduction

Following [1], for $f : \mathbb{R}^p \rightarrow \mathbb{R}$ we consider the optimisation problem

$$\min_{x \in \mathbb{R}^p} f(x) \tag{1}$$

where f is continuous, L -smooth and convex. We write

$$f_\star = f(x_\star), \quad \forall x_\star \in \operatorname{argmin}_{x \in \mathbb{R}^p} f. \tag{2}$$

Definition 1. (Nesterov Accelerated Gradient Descent)

Let $x_0 = y_0 \in \mathbb{R}^p$. Nesterov Accelerated Gradient Descent (NAG) is defined by the iterative construction of the following sequences:

$$\begin{aligned}
x_{k+1} &= y_k - \frac{1}{L} \nabla f(y_k) \\
y_{k+1} &= x_k + \frac{t_k - 1}{t_{k+1}} (x_{k+1} - x_k) \\
k &= 0, 1, \dots
\end{aligned} \tag{3}$$

where $\{t_k\}$ satisfies

$$t_0 := 1, \quad t_{k+1}^2 - t_{k+1} \leq t_k^2. \tag{4}$$

Reformulation

This lemma rewrites (3) in terms of an intermediate sequence $\{z_k\}$.

Lemma 2. (Reformulation)

The following iterative construction of $\{x_k\}$ and $\{y_k\}$ produces the same sequences as NAG from (3).

$$x_{k+1} = y_k - \frac{1}{L} \nabla f(y_k), \tag{5}$$

$$z_{k+1} = z_k - \frac{t_k}{L} \nabla f(y_k), \tag{6}$$

$$y_{k+1} = \left(1 - \frac{1}{t_{k+1}}\right) x_{k+1} + \frac{1}{t_{k+1}} z_{k+1}, \tag{7}$$

where $x_0 = y_0 = z_0 \in \mathbb{R}^p$ and $\{t_k\}$ is as in (4).

Proof. The proof is direct by algebraic manipulations. Note that only the definition of $\{y_k\}$ is different to that of [Definition 1](#), so it only remains to show (7) satisfies (3).

Using (5) in (6) gives

$$z_{k+1} = z_k - t_k (y_{k+1} - x_{k+1}). \tag{8}$$

Rearranging (7) gives

$$t_{k+1} y_{k+1} - (t_{k+1} - 1) x_{k+1} = z_{k+1} \tag{9}$$

which, evaluated at step k yields

$$t_k y_k - (t_k - 1) x_k = z_k. \tag{10}$$

Substituting (10) into (8) gives

$$\begin{aligned}
z_{k+1} &= z_k - t_k(y_{k+1} - x_{k+1}) \\
&= t_k x_{k+1} - (t_k - 1)x_k.
\end{aligned} \tag{11}$$

Finally, using (11) in (7) yields the result:

$$\begin{aligned}
y_{k+1} &= \left(1 - \frac{1}{t_{k+1}}\right)x_{k+1} + \frac{1}{t_{k+1}}(t_k x_{k+1} - (t_k - 1)x_k) \\
&= x_{k+1} + x_{k+1} \left(\frac{t_k}{t_{k+1}} - \frac{1}{t_{k+1}}\right) - x_k \left(\frac{t_k - 1}{t_{k+1}}\right) \\
&= x_{k+1} + \frac{t_k - 1}{t_{k+1}}(x_{k+1} - x_k).
\end{aligned} \tag{12}$$

■

This lemma provides some intuition for how NAG works. Due to the factor of t_k , at each iteration z_k steps farther than a normal gradient step, accelerating the early progress but potentially overshooting. At the same time, the ‘normal’ iteration y_k is a weighted average of x_k and z_k that places an increasing weight on x_k with each iteration, and all the weight in the limit. Somehow it turns out that these two processes— $\{z_k\}$ taking larger and larger steps but having less and less influence on $\{y_k\}$ —balance each other perfectly to accelerate the convergence to $O(1/k^2)$.

Potential

[1] uses a *potential function* method to prove point convergence. Loosely, a potential function (key references: [2], [3]) is a function of the form defined below.

Apparently, the standard way to view these functions is as ‘Lyapunov functions’. I believe that perspective is more relevant for continuous time treatments.

Definition 3. (Potential Function) A potential function $\Phi_k(x) : \mathbb{R}^p \rightarrow \mathbb{R}_+$, usually written as Φ_k since it is usually only evaluated at x_* , is defined as

$$\Phi_k = a_k(f(x_k) - f(x_*)) + b_k\|x_k - x_*\|, \quad k \in \mathbb{N}^* \tag{13}$$

for $a_k, b_k \geq 0$.

The whole idea of using these functions is to prove $\Phi_{k+1} - \Phi_k \leq B_k$, for some sequence $\{B_k\}_{k \geq 0}$ (often $B_k = 0$).

The following lemma is really the core of the whole proof, everything that comes after in the proof is standard analysis.

Lemma 4. (Potential)

Proof. ()

Boundedness

Given [Lemma 2](#) and [Lemma 4](#), this is a relatively simple lemma that ensures that the main sequence of iterates $\{x_k\}$ in NAG is bounded.

Lemma 5. (Boundedness)

Convergence

This lemma is technical in nature and concerns a particular convergence criteria for a real, strictly positive sequence. Before arriving at [Lemma 7](#), the following standard result is needed. For completeness, I provide a statement and proof based on [this link](#).

Theorem 6. (Stolz-Cesáro) If $\{b_k\}_{k \in \mathbb{N}} \subseteq \mathbb{R}_+^*$ is a real, strictly positive sequence such that $\sum_{k=0}^{\infty} b_k = \infty$, then for any sequence $\{a_k\}_{k \in \mathbb{N}} \subseteq \mathbb{R}$ the following two inequalities hold:

$$\limsup_{k \rightarrow \infty} \frac{\sum_{k=0}^{\infty} a_k}{\sum_{k=0}^{\infty} b_k} \leq \limsup_{k \rightarrow \infty} \frac{a_k}{b_k}, \quad (14)$$

$$\liminf_{k \rightarrow \infty} \frac{\sum_{k=0}^{\infty} a_k}{\sum_{k=0}^{\infty} b_k} \geq \liminf_{k \rightarrow \infty} \frac{a_k}{b_k}. \quad (15)$$

In paritucular, if $\{a_k/b_k\}_{k \in \mathbb{N}}$ is a convergent sequence, then the above inequalities force

$$\lim_{k \rightarrow \infty} \frac{\sum_{k=0}^{\infty} a_k}{\sum_{k=0}^{\infty} b_k} = \lim_{k \rightarrow \infty} \frac{a_k}{b_k}. \quad (16)$$

Proof. (Due to G. Nagy) If (14) holds, then (15) holds by taking $a_k \mapsto -a_k$ and using $\sup_{k \geq n} -a_k = -\inf_{k \geq n} a_k$. Therefore it only remains to show (15). If the RHS of (14) is $+\infty$, then the result is trivially true. Assume that $\limsup_{k \rightarrow \infty} a_k/b_k = L$ with L finite or $-\infty$, and let $l > L$ fixed. By definition of $\limsup$,

$$\exists k_* \in \mathbb{N} : \frac{a_k}{b_k} \leq l, \quad \forall k > k_*. \quad (17)$$

Let A_k and B_k be the partial sums of $\{a_k\}$ and $\{b_k\}$. By (17), for all $k > k_*$,

$$\begin{aligned}
A_k &= a_1 + a_2 + \cdots + a_k \\
&\leq (a_1 + \cdots + a_{k_*}) + l (b_{k_*+1} + b_{k_*+2} + \cdots + b_k) \\
&= A_{k_*} + l(B_k - B_{k_*}).
\end{aligned} \tag{18}$$

Since $B_k > 0$, dividing by B_k gives

$$\frac{A_k}{B_k} \leq l + \frac{A_{k_*} - B_{k_*}}{B_k}, \quad \forall k > k_*. \tag{19}$$

Fixing k_* and taking $\limsup_{k \rightarrow \infty}$ on both sides gives

$$\limsup_{k \rightarrow \infty} \frac{A_k}{B_k} \leq l, \quad \forall l > L \tag{20}$$

This is a lowerbound for the set $\{l \in \mathbb{R} : l > L\}$ that has infimum (greatest upper bound) L , hence the result. ■

Lemma 7. (Convergence) Let $\{\varphi_k\}_{k \geq 0}, \{t_k\}_{k \geq 0} \subseteq \mathbb{R}_+^*$ be sequences of real and positive numbers respectively, such that $\sum_{k=0}^{\infty} (1/b_k) = +\infty$. If

$$\lim_{k \rightarrow \infty} a_{k+1} + b_k(a_{k+1} - a_k) \rightarrow L \in \mathbb{R}, \tag{21}$$

then

$$\lim_{k \rightarrow \infty} a_k = L \tag{22}$$

Proof. (From cited article)

For all $k \in \mathbb{N}$ let $\ell_k = t_k(\varphi_{k+1} - \varphi_k)$ and suppose that $\ell_k \rightarrow L \in \mathbb{R}$ as $k \rightarrow \infty$. Consider the positive sequence $\{\mu_k\}$ defined recursively as

$$m_0 := \frac{1}{t_0} \quad \text{and} \quad m_{k+1} := \frac{\mu_k + \mu_k t_k}{t_{k+1}}, \quad \forall k \in \mathbb{N}. \tag{23}$$

The proof of will also use the following consequence of the Bolzano-Weierstrass theorem. ■

Lemma 8. (A bounded sequence is convergent if and only if it has a unique accumulation point)

Let $\{x_k\}_{k \geq 0} \subseteq \mathbb{R}^p$ be a bounded sequence. Then $x \rightarrow \ell$ if and only if ℓ is the unique accumulation point of $\{x_k\}$.

Proof. $[\Rightarrow]$: Since x_k is convergent, there can only be one accumulation point ℓ —otherwise you could find a neighbourhood outside of ℓ that is visited infinitely often, which would contradict $x_k \rightarrow \ell$. $[\Leftarrow]$: Towards a contradiction suppose that $x \not\rightarrow \ell$. Then, there exists $x_{\gamma(k)}$ and $r > 0$ such that $\|x_{\gamma(k)} - \ell\| > r$ for all $k \geq 0$. $\{x_{\gamma(k)}\}$ is a bounded sequence, hence by BW it has a convergent subsequence with an accumulation point different to ℓ .⁴ Hence $x \rightarrow \ell$. ■

Theorem 3.5

The high-level strategy of the proof is as follows.

1. Rewrite NAG using [Lemma 2](#).
2. Use [Lemma 5](#) to conclude that $\{x_k\}$ has at least one convergent subsequence (by Bolzano-Weierstrass). If $\{x_k\}$ has *exactly one* convergent subsequence, then the proof is finished by [Lemma 8](#).
3. Assume there are two accumulation points; the rest of the proof will show that these must coincide.
4. Use [Lemma 4](#) to set up something of the form required by [Lemma 7](#). Verify the hypothesis of [Lemma 7](#) using the definition of $\{t_k\}$ and standard analysis.
5. Conclude that the two accumulation points must coincide.

Theorem 9. (Theorem 3.5 of [1])

Bibliography

- [1] U. Jang and E. K. Ryu, “Point Convergence of Nesterov's Accelerated Gradient Method: An AI-Assisted Proof.” 2026.
- [2] A. d'Aspremont, D. Scieur, and A. Taylor, “Acceleration Methods,” *Foundations and Trends® in Optimization*, vol. 5, no. 1–2, pp. 1–245, Dec. 2021, doi: [10.1561/24000000036](https://doi.org/10.1561/24000000036).
- [3] N. Bansal and A. Gupta, *Theory of Computing*, vol. 15, no. 1, pp. 1–32, 2019, doi: [10.4086/toc.2019.v015a004](https://doi.org/10.4086/toc.2019.v015a004).